

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 \delta r_+ &= \delta \left(M + \sqrt{M^2 - a^2 - Q^2} \right) \\
 &= \delta M + \frac{1}{2} (M^2 - a^2 - Q^2)^{-\frac{1}{2}} 2 (M \delta M - a \delta a) \\
 &= \delta M + (r_+ - M)^{-1} (M \delta M - a \delta a) , \\
 a \delta a &= a \delta (J/M) = \frac{a}{M} \delta J - \frac{a^2}{M} \delta M .
 \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 A \circ_X B &= (AX)(X^\dagger B) \\
 &= (AX^{2/3}X^{1/3})(X^{-1/3}X^{-2/3}B) \\
 &= X^{1/3}[(X^{-1/3}AX^{2/3})(X^{-2/3}B)] \\
 &= X^{1/3}[(X^{-1/3}AX^{1/3}X^{1/3})(X^{-1/3}X^{-1/3}B)] \\
 &= X^{1/3}\{[(X^{-1/3}AX^{1/3})(X^{-1/3}BX^{1/3})]X^{-1/3}\} \\
 &= X^{1/3}[(X^{-1/3}AX^{1/3})(X^{-1/3}BX^{1/3})]X^{-1/3}.
 \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$F_{p;r}^{(L)}(q) = q^{-r(r-1)} \times \begin{cases} F_{p-2}^{(L)} \begin{bmatrix} \mathbf{Q}_{r,1} \\ \mathbf{0} \end{bmatrix} (\mathbf{e}_r|q) & \text{if } L \not\equiv r-1 \pmod{2}, \\ F_{p-2}^{(L)} \begin{bmatrix} \mathbf{Q}_{p-r,p} \\ \mathbf{0} \end{bmatrix} (\mathbf{e}_{p-r}|q) & \text{if } L \equiv r-1 \pmod{2}, \end{cases}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 u(\tau) &= e^{\sqrt{\gamma^2 + m^2/2} \tau} \left[u_0 \cosh(\gamma \tau) - \left(u_0 \sqrt{\gamma^2 + m^2/2} - \dot{u}_0 \right) \frac{1}{\gamma} \sinh(\gamma \tau) \right] \\
 v(\tau) &= e^{-\sqrt{\gamma^2 + m^2/2} \tau} \left[v_0 \cosh(\gamma \tau) + \left(v_0 \sqrt{\gamma^2 + m^2/2} + \dot{v}_0 \right) \frac{1}{\gamma} \sinh(\gamma \tau) \right],
 \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$${}^*f_{\mu\nu}(x) = -\frac{e_0}{4\pi\epsilon_0 c}[\partial_\mu\vec{n}(x) \wedge \partial_\nu\vec{n}(x)]\vec{n}(x) = \begin{pmatrix} 0 & B_x & B_y & B_z \\ -B_x & 0 & \frac{E_z}{c} & \frac{-E_y}{c} \\ -B_y & \frac{-E_z}{c} & 0 & \frac{E_x}{c} \\ -B_z & \frac{E_y}{c} & \frac{-E_x}{c} & 0 \end{pmatrix}$$